

SBC 201 Chapter 7 Fire and Smoke Protection Features — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–20 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 7, source file Reference\SBC 201 2024\source_reference\Chapter_07 – FIRE AND SMOKE PROTECTION FEATURES.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **2863** independently checkable numeric records (**517** Verified, **2346** Verify source). The Verify-source majority is the OCR-destroyed Tables 721.1(1)–(3) and 722.* cell dump; those cells are not design-release values.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, fire-engineering report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Tables 601 / 508.4 / 509.1 / 1020.2, Sections 403 / 404 / 414 / 420 / 510 / 1023 / 1024 / 1026, Chapter 9 sprinkler design, Chapter 10 corridor ratings, Chapter 14 cladding, NFPA 82, SBC 501 plenum limits, SBC 801 ESS enclosure ratings, commentary examples, or a chapter summary. Where Chapter 7 sends the user elsewhere, this matrix records the dependency without supplying the outbound value. No Chapter 7 *_CS.md was found.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Construction type is unconfirmed. A high-rise R-2 tower is typically Type I or II, not verified. Type V framed walls are not assumed for the tower shaft.
4. Automatic sprinkler protection is not selected. **NFPA 13 / Section 903.3.1.1** and **NFPA 13R / Section 903.3.1.2** are shown as separate branches. Table 705.8 UP, S is **903.3.1.1 only**.
5. Fire-separation distance by elevation, fire walls versus occupancy/fire-area barriers, podium mix, adjoining S-2 parking, storey count and shaft program are unconfirmed.
6. Tables 716.1(2) (flattened remainder), 716.1(3) (>ch1 / exterior 1/2 token), Figure 705.7 F_eo, and all appended Tables 721.1(1)–(3) and 722.* are OCR-unresolved. Affected rows are **Verify source**.
7. Exterior-wall weather envelope and combustible cladding remain in Chapter 14. Structural FRR by construction type remains in Table 601. High-rise SFRM bond strength and the 403.2.1.2 shaft-hour reduction remain in Section 403.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, construction type, sprinkler branch, FSD, mixed-use or exception exists.
Not typical	Unrelated occupancy-only or low-rise-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 7 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 7 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, missing figure, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 7 application	Required project action
Construction type	Unconfirmed; high-rise R-2 typically Type I or II, not verified	Table 705.5 Type I vs IIB/VB bands; 704 encasement; 708.3 / 711.2.4.3 half-hour unit-separation exception is Type IIB/IIIB/VB only; 718 combustible fireblocking is Type III–V	Freeze type of construction on the code data sheet before wall, shaft and SFRM schedules
Sprinkler basis	Unconfirmed: neither NFPA 13 nor 13R is assumed	Table 705.8 UP, S is 903.3.1.1 only ; 705.8.5 vertical opening skip allows 13 or 13R; 708.3 Exc 2 / 711.2.4.3 Exc are 903.3.1.1 only ; 717.5.3 Exc 2 smoke-damper skip is 903.3.1.1	Fire engineer to lock Chapter 9 system; do not treat 13R as the Table 705.8 sprinklered opening branch
Fire-separation distance	Unconfirmed by elevation	Tables 705.2 / 705.5 / 705.8; 705.5 both-sides rating at FSD ≤ 3 m ; 705.2.3 projection protection within 1500 mm	Dimension FSD on each elevation to the lot line or assumed lot line, not to the balcony edge
Storeys and height	Occupied floor stated above 23 m; exact storey count unconfirmed	Shafts connecting 4 or more stories are 2 h (713.4); 705.8.5 Exc 1 is ≤ 3 stories; 708.4.2 Exc 4 R-2 draftstop reduction is ≤ 4 stories and ≤ 18 m	Issue a signed code datum sheet with storey count including basements, excluding mezzanines
Fire walls vs barriers	Unconfirmed whether a fire wall, occupancy separation or fire-area barrier is used	Table 706.4 R-2 fire wall 3 h (2 h in Type II/V); Table 707.3.10 R fire area 2 h ; Table 508.4 mixed-occupancy hours not imported	Freeze the separation strategy on the life-safety plans; do not treat a fire barrier as a fire wall
Podium / mixed use / S-2 garage	Unconfirmed	705.3 Exc 2 adjoining S-2/R-2 openings; 707.3.9 mixed occupancy; incidental uses Table 509.1	Classify podium, parking and amenity occupancies; tag shared walls with all served groups
High-rise Section 403	Unconfirmed beyond the 23 m project statement	Commentary (not code) cites 403.2.1.2 shaft-hour reduction and 403.2.3 SFRM bond; neither value is in Chapter 7	Fire engineer to apply Section 403 separately; do not adopt 1-hour shaft reductions from commentary
Corridor rating	Unconfirmed	708.1 / 708.3 send corridor walls to Table 1020.2; ½-hour corridor option is outbound	Use the Chapter 10 matrix for corridor hours; keep 708 construction rules here
Incidental uses	Unconfirmed (trash, linen, parking attendant, etc.)	707.3.7 / 713.13.5 send hours to Table 509.1	Schedule incidental rooms; do not import 509.1 hours into this matrix

Decision / gap	Current project basis	Why it controls Chapter 7 application	Required project action
NOC and fire strategy	SCD NOC and stamped fire-strategy status unconfirmed	Shaft pressurization, smoke control, damper exceptions and SCD acceptance cannot be concluded from Chapter 7 alone	Engage the qualified local/fire consultant before design freeze

4. Fire tests, identification and multiple-use assemblies

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
702.1	Multiple-use fire assemblies	An assembly serving more than one fire purpose shall comply with all applicable requirements for each purpose	Wall, floor, door or damper serving two or more listed roles	None stated	Direct	Tag corridor/shaft/lobby walls with every applicable 707/708/709/716/717 rule	Verified
703.2	Fire-resistance without sprinklers	FRR of elements, components or assemblies is determined by 703.2.1 or 703.2.2 without automatic sprinklers or other suppression in the test, or by 703.2.3	Required fire-resistance rating	Alternative method under 104.11	Direct	Specify tested/analytical listings; do not take credit for building sprinklers in the E119/UL 263 rating	Verified
703.2.1.1	Nonsymmetrical interior walls	Test both faces; assigned rating is the shortest duration	Interior walls and partitions of nonsymmetrical construction	One-face test only if evidence shows the least-resistant face was exposed (building official acceptance); exterior walls follow 705.5	Direct	Use symmetrical shaft/corridor assemblies or document the weaker-face listing	Verified
703.3.1 Exc	Composite noncombustible surfacing	Structural base noncombustible (ASTM E136 or E2652 with E136 criteria) plus surfacing not more than 3.18 mm with flame-spread index not greater than 50 (ASTM E84 or UL 723)	Materials required to be noncombustible	None stated	Direct	Keep thin finishes on noncombustible substrates within the thickness and FSI limits	Verified
703.5	Rated-wall marking in concealed space	Permanent signs or stenciling within 4.5 m of each wall end and at intervals not exceeding 9 m ; lettering not less than 75 mm high with minimum 10 mm stroke	Accessible concealed floor, floor-ceiling or attic at fire walls, fire barriers, fire partitions, smoke barriers, smoke partitions or other walls requiring protected openings	None stated	Direct	Show barrier identification on reflected-ceiling and attic plans	Verified
703.6	Mass-timber protection time	Contribution minutes = Test Assembly 2 time minus unprotected Test Assembly 1 time, same ASTM E119 / UL 263 structural-failure criteria	Types IV-A / IV-B / IV-C noncombustible protection	Table 722.7.1(1) deemed-to-comply times are Verify source	Conditional	Do not use Type IV unless mass timber is specified; verify published 722.7 tables before assigning minutes	Verified

5. Structural member protection and SFRM

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
704.1	Structural FRR floor	Member FRR not less than the rated assemblies it supports, and not less than Table 601	Columns, beams and other required-rated members	Fire barriers, fire partitions, smoke barriers and horizontal assemblies as provided in 707.5, 708.4, 709.4 and 711.2	External verification	Take hours from Table 601 and from the supported assembly; do not import Table 601 here	Verified
704.2	Column individual encasement	Protect the entire column on all sides for the full height, including connections, through the ceiling space to the top of the column	Columns required to have protection to achieve an FRR	Light-frame integral studs/columns per 704.4.1	Direct	Continue SFRM or board encasement through the ceiling plenum; do not stop at the ceiling membrane	Verified
704.3	Primary-frame individual encasement	Individual encasement on all sides for the full length, including connections	Primary frame other than columns that supports more than two floors , or one floor and a roof , or a load-bearing wall, or a nonload-bearing wall more than two stories high	Exposed-sides-only where the tested extent satisfies 703	Direct	Encase transfer girders and multi-storey primary beams individually, not by floor membrane alone	Verified
704.6–704.6.1	Attachment remaining cover	Lug/bracket/rivet/bolt edges may extend to within 25 mm of the fire-protection surface; secondary steel attachments receive the same material and thickness for not less than 300 mm , or the full length if shorter than 300 mm ; hollow open-end attachments protected inside and out	Primary or secondary structural steel requiring fire protection	None stated	Direct	Extend SFRM onto hangers, braces and erection lugs; close open hollow stubs	Verified
704.7	Reinforcement cover measurement	Protection thickness measured to the outside of reinforcement; stirrups and spiral ties may project not more than 12.5 mm into the protection	Concrete or masonry members with required cover	None stated	Direct	Check cover to main bars, not to stirrup outer face	Verified
704.9	Impact protection height	Corner guards or a substantial noncombustible jacket to a height adequate for full protection, not less than 1.5 m above the finished floor	Fire-protective covering subject to vehicle, merchandise or other impact	Concrete columns in parking garages	Conditional	Protect SFRM columns in the podium garage and loading bay to at least 1.5 m	Verified
704.10	Exterior load-bearing member FRR	Highest of Table 601 for the element, Table 601 for exterior bearing walls, and Table 705.5 based on FSD	Load-bearing members in exterior walls or outside the building	None stated	External verification	Coordinate column FRR with FSD and construction type; do not import Table 601 hours	Verified
704.11	Unprotected lintel bottom flange	Fire protection not required at the bottom flange of lintels, shelf angles and plates spanning not more than 2 m (primary frame or not), and at bottom flanges not part of the structural frame regardless of span	Lintels, shelf angles and plates	None stated	Conditional	Leave short masonry lintel soffits unprotected only within the 2 m primary-frame limit	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
704.13.3.2	SFRM over unlisted primer	Field ASTM E736: not fewer than five bond tests; average bond not less than 80% and individual not less than 50% of SFRM on clean uncoated 3.2 mm plate; beam flange ≤ 300 mm , column flange ≤ 400 mm , web depth ≤ 400 mm	SFRM over primers, paints or encapsulants other than those in the listing	Do not use this path outside the size limits	Conditional	Prefer listed primers; if painted steel is unavoidable, run the five-test E736 protocol before spraying	Verified
704.13.4	SFRM application temperature	Ambient and substrate not less than 4.5°C during application and for not fewer than 24 hours after	SFRM application	Manufacturer's instructions may allow otherwise	Direct	Hold winter/night pours and spray until the 4.5°C / 24 h condition is met	Verified
704.13 / 403.2.3	High-rise SFRM bond	Chapter 7 requires listing-consistent thickness, density and substrate; bond-strength values are in Section 403.2.3 , not here	High-rise steel frame with SFRM	None stated in Chapter 7	External verification	Specify 1705.14 special inspection and Section 403 bond; do not invent kPa values	Verified

6. Exterior walls — projections, FSD ratings and openings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Table 705.2	Projection setback from FSD line	FSD 0 to < 600 mm : projections not permitted ; 600 to < 900 mm : ≥ 600 mm ; 900 to < 1500 mm : ≥ two-thirds of FSD ; ≥ 1500 mm : ≥ 1000 mm	Cornices, eaves, balconies and similar projections	705.3 same-lot buildings treated as one building	Direct	Measure FSD to the wall, then check balcony/eave edge against Table 705.2	Verified
705.2.3	Projection protection near lot line	Projections within 1500 mm of the FSD line shall be noncombustible, ≥ 1-hour combustible, heavy timber 2304.11, FRTW, or 705.2.3.1	Projections extending toward the FSD line	Type VB Group R-3/U with FSD ≥ 1.5 m (not this tower)	Conditional	Fire-rate or make noncombustible any balcony/eave that enters the 1500 mm zone	Verified
705.2.3.1	Combustible balcony length	Aggregate length not more than 50 percent of the building perimeter on each floor; combustible construction other than FRTW shall match Table 601 floor FRR or be heavy timber	Combustible balconies and similar projections	Exc 4: sprinklers extended to the balcony → length not limited. Exc 2: untreated wood/plastic-composite pickets ≤ 1050 mm . Exc 1: FRTW on Types I/II ≤ 3 stories (not this tower)	Conditional	Either sprinkler every balcony or keep combustible balcony length ≤ 50% of that floor perimeter	Verified
705.3 Exc 2	Adjoining S-2 / R-2 openings	Where an S-2 parking garage of Type I or IIA has no FSD to a Group R-2 building, occupant-use openings are permitted per 706.8; opening protectives not less than 1½ hour are required only in the S-2 wall , not in the R-2 wall	S-2 garage erected on the same lot as Group R-2 with zero FSD	Protectives only on the garage side	Conditional	If podium parking shares a wall with the tower, rate the garage openings 1½ h and keep the residential openings unprotected on that interface	Verified
705.5	Exterior-wall fire exposure	Walls with FSD ≤ 3 m rated from both sides ; FSD > 3 m rated from the inside only	Exterior walls required to have an FRR	None stated	Direct	Use two-sided listings on close lot-line walls; interior-only listings only where FSD exceeds 3 m	Verified
Table 705.5	Group R exterior-wall FRR	Group R column: FSD < 1.5 m all types 1 h ; 1.5 ≤ FSD < 3 m 1 h ; 3 ≤ FSD < 9 m Type IA/IB/IVA/IVB 1 h (note c), Type IIB/VB 0 h , other types 1 h ; FSD ≥ 9 m all types 0 h . Load-bearing walls also comply with Table 601 (not imported). Note g: 0 h where Table 705.8 permits unlimited unprotected openings on nonbearing walls	Exterior walls, FSD measured to the story	Notes b, c, g, h, i; Group H / F-1 / M / S-1 columns omitted from this row	Direct	For a Type I tower, schedule 1 h exterior walls where FSD < 9 m , then overlay Table 601 if the wall is load-bearing	Verified
Table 705.8	Exterior opening area per story	Percentage of exterior-wall area per story. UP, NS / UP, S (903.3.1.1 only) / Protected: FSD 0–<0.9 m Not Permitted / Not Permitted / Not Permitted; 0.9–<1.5 m NP / 15% / 15% ; 1.5–<3 m 10% / 25% / 25% ; 3–<4.5 m 15% / 45% / 45% ; 4.5–<6 m 25% / 75% / 75% ; 6–<7.5 m 45% / No Limit / No Limit ; 7.5–<9 m 70% / No Limit / No Limit ; ≥ 9 m No Limit all degrees	Exterior wall openings, each story	Note k → 705.3 Exc 2 for S-2/R-2. Notes d/f are Group R-3 only. Note g: open parking FSD ≥ 3 m unlimited. Do not treat 13R as UP, S	Direct	Calculate opening % by FSD and story; until NFPA 13 is locked, use the UP, NS column as the conservative branch	Verified
705.8.1 Exc 1	First-story unlimited unprotected	Street-facing first story with FSD > 4.5 m to street centreline, or facing ≥ 9 m unoccupied space with fire-lane access	First story above grade plane; other than Group H	Does not unlock upper-storey openings	Conditional	Apply only to the ground-storey street elevation	Verified
705.8.4	Mixed protected and unprotected openings	$((A_p/ap)+(A_u/au) \leq 1)$ (Equation 7-2)	Both opening types in the same story exterior wall	(A_e) from 705.7 may replace (A_p)	Direct	Do not add protected and unprotected percentages independently	Verified
Figure 705.7	Equivalent opening factor (F_{eo})	Figure lookup required for Equation 7-1; no numeric table in the extract	Protected openings where unexposed-temperature criterion is waived	Do not guess (F_{eo})	External verification	Verify the published figure before using equivalent protected area	Verify source
705.8.5	Adjacent-story opening separation	Where openings in adjacent stories are within 1.5 m horizontally and the lower opening is not a ≥ ¾-hour protective: provide ≥ 900 mm of ≥ 1-hour both-sides spandrel, or a ≥ 1-hour flame barrier projecting ≥ 750 mm	Exterior openings in adjacent stories	Exc 1: ≤ 3 stories. Exc 2: sprinklers throughout 903.3.1.1 or 903.3.1.2 . Exc 3: open parking	Conditional	If 13 or 13R is locked throughout, document Exc 2 and omit the spandrel; otherwise detail the 900 mm / 750 mm option	Verified
705.8.6	Same-lot vertical roof exposure	Opening < 4.5 m vertically above an adjacent roof, with FSD to the imaginary line < 4.5 m each side, shall be a ≥ ¾-hour protective	Buildings on the same lot	Exc 1: ≥ 1-hour roof for ≥ 3 m from the facing wall, including supports	Conditional	Protect low podium-roof windows facing a higher tower wall, or rate the lower roof	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
705.11 / 705.11.1	Parapet height	Required parapets ≥ 750 mm above the roof/wall intersection; uppermost 450 mm on the roof side noncombustible including counterflashing/coping. Roof slope > 2:12 (16.7-percent) toward the parapet: extend to the height of any roof within an FSD that requires opening protection	Parapet required by 705.11	Exc 4: 1-hour wall to underside of roof with Group R parallel framing ≥ 1.2 m and roof openings ≥ 1.5 m . Exc 6: Table 705.8 permits ≥ 25% unprotected. Exc 2: floor area ≤ 100 m²	Conditional	Provide 750 mm parapets on close FSD walls unless a numbered exception is demonstrated	Verified

7. Fire walls

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Table 706.4	R-2 fire-wall rating	Groups A, B, E, H-4, I, R-1, R-2, U: 3 hours . Note a: Type II or V walls permitted to be 2 hours	Fire wall used to create separate buildings	H-1/H-2 4 h ; F-2/S-2/R-3/R-4 2 h (not this occupancy row)	Conditional	If a fire wall is used, schedule 3 h unless both sides are Type II or V	Verified
706.5	Horizontal fire-wall extension	Extend ≥ 450 mm beyond the exterior surface	Fire wall, exterior wall to exterior wall	Exc 1–2: terminate at interior of sheathing with ≥ 1.2 m of ≥ 1-hour exterior each side and ≥ ¾-hour openings	Conditional	Carry the fire wall through the facade or document the 1.2 m rated-skin exception	Verified
706.5.1	Exterior walls at fire-wall intersection	1-hour exterior walls with ¾-hour openings for ≥ 1.2 m each side, unless the angle is ≥ 180 degrees	Fire wall meets an exterior wall	Imaginary-lot-line protection also skipped at ≥ 180 degrees	Conditional	Detail the re-entrant corner or prove a straight 180-degree facade	Verified
706.5.2	Projecting elements near fire wall	Fire wall extends to the outer edge of any balcony, overhang, canopy, marquee or similar projection ≤ 1.2 m from the wall	Horizontal projections beside a fire wall	Listed 1-hour / ¾-hour opening exceptions 1–3	Conditional	Stop balconies short of 1.2 m or extend the fire wall through the projection	Verified
706.6	Fire wall above roofs	Terminate ≥ 750 mm above both adjacent roofs	Fire walls	Exc 2: 2-hour wall may stop at roof deck with 1-hour roof within 1.2 m and openings ≥ 1.2 m . Exc 5: walls above a 3-hour 510.2 podium assembly may start at the top of that assembly (510 hours not imported)	Conditional	Parapet the fire wall 750 mm unless a numbered roof-termination exception is proven	Verified
706.6.1	Stepped-building fire wall	Terminate ≥ 750 mm above the lower roof; exterior wall extending > 750 mm above the lower roof is ≥ 1 hour both sides with ≥ ¾-hour openings; portion > 4.5 m above the lower roof need not be rated unless other provisions apply	Fire wall also serving as exterior wall at different roof levels	Exception: 1-hour lower roof within 3 m , supports 1 hour , no openings within 3 m	Conditional	Section the podium/tower step and apply the 750 mm / 4.5 m stations	Verified
706.6.2	Sloped roof at interior fire wall	Where slope toward the fire wall is > 2:12 , wall height equals roof height at 1200 mm from the wall plus 750 mm , and not less than 750 mm	Interior fire wall with sloping roofs	None stated	Conditional	Raise the fire-wall parapet on the uphill side of a sloped roof	Verified
706.7	Combustible members into masonry fire wall	Opposite-side embedded ends ≥ 100 mm apart; hollow walls filled solid ≥ 100 mm above, below and between members	Combustible members bearing into a concrete or masonry fire wall	None stated	Conditional	Offset timber/joist pockets through the fire wall	Verified
706.8	Fire-wall openings	Each opening ≤ 15 m² ; aggregate width ≤ 25% of the wall length at any floor	Openings through a fire wall	Exc 1: party walls — no openings. Exc 2: both buildings sprinklered 903.3.1 throughout — 15 m² not limited; 25% still applies	Conditional	Size fire-wall doors to the 15 m² / 25% cap	Verified

8. Fire barriers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
707.3.1 / 713.4	Shaft barrier rating	Use 713.4 (in this chapter): ≥ 2 h connecting 4 or more stories; ≥ 1 h connecting fewer than 4; not less than the floor penetrated; need not exceed 2 h	Fire barrier enclosing a shaft	Interior exit stairs follow 1023.1 (hours not imported)	Direct	Schedule 2-hour shaft walls for the tower; do not import 403.2.1.2 reductions from commentary	Verified
707.3.2–707.3.5	Exit-enclosure ratings	Interior exit stair/ramp → 1023.1; exit passageway → 1024.3; horizontal exit → 1026.1; exit-access stair enclosure → 713.4	Fire barriers for those exits	Values not filled from Chapter 10	External verification	Coordinate exit-enclosure hours with the Chapter 10 matrix	Verified
707.3.7	Incidental-use barrier	FRR not less than Table 509.1	Incidental uses	Sprinkler-in-lieu walls that only resist smoke are not fire barriers	External verification	Do not import Table 509.1 hours; list trash, linen and similar rooms on the incidental-use schedule	Verified
707.3.9	Mixed-occupancy barrier	FRR not less than Table 508.4	Separated mixed occupancies	Also apply 707.3.10 if the wall separates fire areas	External verification	Do not import Table 508.4; tag podium retail/parking separations for a Chapter 5 check	Verified
Table 707.3.10	Fire-area separation rating	Group R (with A, B, E, F-2, H-4, H-5, I, M, S-2); 2 hours . Mixed-occupancy fire areas: highest table value. H-1/H-2 4 h ; F-1/H-3/S-1 3 h ; U 1 h	Fire barriers, fire walls or horizontals separating fire areas	Do not use this table as occupancy-separation (that is Table 508.4)	Direct	If fire areas are used, separate R fire areas with 2-hour barriers or horizontals	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
707.5 / 707.5.1	Fire-barrier continuity and support	Continuous from foundation or floor/ceiling below through concealed space to the underside of the floor/roof sheathing, deck or slab; supporting construction FRR ≥ the barrier; hollow vertical spaces fireblocked at every floor	Fire barriers	Incidental-use ≤ 1 h support unrated in Type IIB/IIIB/VB (Table 509.1 hours not filled). Exit-passageway top may match 1024.3	Direct	Run shaft and occupancy-separation walls through the ceiling; fireblock stud cavities at each floor	Verified
707.6	Fire-barrier opening limits	Aggregate opening width ≤ 25% of wall length; each opening ≤ 15 m²	Openings in a fire barrier	Exc 1: 15 m² waived where adjoining floors have 903.3.1.1 (25% remains). Exc 2: both limits waived for fire doors serving exit-access or interior-exit stair/ramp enclosures. Exc 3: E119/UL 263 protective with FRR ≥ wall. Exc 4: 25% waived for atrium fire windows. Exc 5: both waived for stair-to-passageway doors	Direct	Keep incidental and occupancy-separation openings within 25% / 15 m² unless a numbered exception applies	Verified

9. Fire partitions — dwelling units and corridors

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
708.1	Fire-partition applications	Includes Group R unit separations (420.2), corridor walls (1020.3), elevator lobby (3006.3), egress balconies (1021.2), R-1/R-2 dwelling/sleeping separations (907.2.8.1 / 907.2.9.1), vestibules (1028.2)	Walls listed in 708.1	Mall tenant and ambulatory walls omitted unless podium mix is opened	Direct	Classify unit demising, corridor and lobby walls as fire partitions, not fire barriers	Verified
708.3	Fire-partition rating	Not less than 1 hour	Fire partitions	Exc 1: corridor ½ hour only where Table 1020.2 permits (table not imported). Exc 2: DU/SU separations ≥ ½ hour in Type IIB/IIIB/VB with 903.3.1.1 throughout (not 13R)	Direct	Schedule 1-hour unit and corridor partitions for a Type I/II tower; do not use the ½-hour unit exception on Type I	Verified
708.4	Fire-partition continuity	From top of foundation or floor/ceiling below to underside of sheathing/deck/slab or to a floor/roof-ceiling assembly with FRR ≥ the partition	Fire partitions	Exc 1: crawl with floor above ≥ 1 h . Exc 2: corridor wall may stop at corridor ceiling membrane if membrane matches the wall and (2.1) room-side membrane continues, or (2.2) 13 or 13R throughout including the interstitial . Exc 3: tunnel corridor with rated ceiling	Direct	Prefer deck-to-deck unit walls; if using a corridor membrane termination, document Exc 2.1 or 2.2	Verified
708.4.1	Supporting construction	FRR ≥ the supported partition	Fire-partition support	Type IIB/IIIB/VB: support rating not required for DU/SU, corridors, mall tenants, ambulatory, R-1/R-2 unit walls, vestibules	Direct	On Type I/II, keep floors supporting 1-hour partitions at least 1 hour (Table 601 still governs)	Verified
708.4.2	Combustible interstitial fireblock/draftstop	Where partitions do not reach the deck, fireblock (718.2.1) or draftstop (718.3.1 / 718.4.1) on the partition line	Combustible construction	Exc 1: 903.3.1.1 throughout, or 903.3.1.2 with interstitial protection as for 13. Exc 2: corridor that also separates units — draftstop above one wall only. Exc 3: R-2 < 4 dwelling units. Exc 4: R-2 ≤ 4 stories and ≤ 18 m — attic draftstop ≤ 280 m² or every 2 units, whichever smaller	Conditional	High-rise exceeds Exc 3–4; if combustible concealed space exists, fireblock/draftstop unless Exc 1 is proven	Verified

10. Smoke barriers and smoke partitions

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
709.3	Smoke-barrier rating	1-hour fire-resistance rating	Smoke barriers	Group I-3 ≥ 2.5 mm steel (not this occupancy)	Conditional	Rate AOR, elevator-lobby and 909 smoke-barrier walls 1 hour when those features exist	Verified
709.4	Smoke-barrier continuity	Effective membrane from foundation or floor/ceiling below to underside of sheathing/deck/slab, including concealed and interstitial spaces; supporting construction rated except Type IIB/IIIB/VB	Smoke barriers	Interstitial exception where ceiling/exterior wall provides equivalent fire and smoke resistance. Horizontal continuity: 709.4.1 compartments vs 709.4.2 AOR/lobby	Conditional	Run smoke barriers through the ceiling void; do not stop at a lay-in ceiling unless the exception is demonstrated	Verified
709.4.2	AOR / elevator-lobby enclosure stop	Terminate at a ≥ 1-hour fire barrier, another smoke barrier, or an outside wall	Smoke barriers enclosing an area of refuge (1009.6.4) or elevator lobby (405.4.3, 3007.6.2, 3008.6.2)	Smoke/draft door not required at the hoistway door or the AOR-to-exit door	Conditional	Close the lobby/AOR smoke enclosure to a rated barrier or the exterior	Verified
710.5.2.2	Smoke-partition door leakage	≤ 0.015 m³/s per m² of door at 25 Pa , ambient and elevated temperature (UL 1784; NFPA 105 installation)	Where smoke-and-draft-control doors are required in smoke partitions	None stated	Conditional	Specify UL 1784 doors if I-2-style smoke partitions or 710 doors appear	Verified

I-2/I-3 smoke-compartment walls, 20 mm cross-corridor undercuts and I-2 pass-throughs are omitted (Not typical).

11. Horizontal assemblies

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
711.2.3	Horizontal supporting construction	Protect supporting construction to the FRR of the horizontal assembly	Rated floors and roofs	Type IIB/IIIB/VB: incidental ≤ 1 h , DU/SU 420.3, and 709 smoke-barrier horizontals may have unrated support	Direct	Encase primary members supporting rated floors (see also 704.3)	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
711.2.4	Floor/roof FRR floor	Not less than type of construction (Table 601, not imported), and not less than 711.2.4.1–711.2.4.6	Horizontal assemblies	Mixed occupancy → 508.4; fire area → Table 707.3.10; incidental → 509	External verification	Overlay Table 601 hours with unit-separation and fire-area hours	Verified
711.2.4.3	Dwelling/sleeping-unit floor rating	Horizontal assemblies serving as DU/SU separations per 420.3: not less than 1 hour	Floors between dwelling or sleeping units	≥ ½ hour in Type IIB/IIIB/VB with 903.3.1.1 throughout	Direct	Keep unit-to-unit floors 1 hour on Type I/II; do not use the ½-hour exception	Verified
711.2.5	Lay-in ceiling uplift	Lay-in panels in a rated floor/roof-ceiling shall resist 48 Pa upward, or hold-downs shall be installed	Lay-in acoustical panels in rated assemblies	None stated	Conditional	Specify hold-down clips or a tested 48 Pa panel	Verified
711.2.6	Unusable crawl or attic membrane	1-hour floor/ceiling over an unusable crawl; ceiling membrane may be omitted. 1-hour roof over an unusable attic; floor membrane may be omitted	Unusable crawl or attic	Usable storage attics are not this exception	Conditional	Omit the unused membrane only where the space is documented as unusable	Verified

12. Vertical openings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
712.1.2	In-unit unconcealed opening	Opening ≤ 4 stories, totally within one dwelling unit, unconcealed	Private unit stair or void	None stated	Conditional	Limit duplex/triplex voids to 4 stories and keep them unconcealed	Verified
712.1.3.1–712.1.3.2	Escalator opening	Opening ≤ 2× escalator projected area; ≤ 4 stories other than Groups B and M; 903.3.1.1 plus draft curtain and NFPA 13 sprinklers. Alternative: ≥ 1.5-hour noncombustible shutters, closing ≤ 150 mm/s , smoke-detector actuated	Escalators penetrating floors	B/M storey cap does not apply	Conditional	If an amenity escalator is added, use the draft-curtain path or listed 1.5 h shutters	Verified
712.1.7	Atrium story threshold	Other than I-2/I-3: atrium where the floor opening connects 3 or more stories (Section 404). I-2/I-3: 2 or more stories	Vertical opening treated as an atrium	A-1/A-4/A-5 balconies and 505 mezzanines are not a story	Conditional	If a lobby void connects 3+ stories, apply Section 404; do not import 404.6 hours	Verified
712.1.9	Two-story opening	Opening connecting not more than 2 stories, other than I-2/I-3, not otherwise listed	Unenclosed two-story void	Not through a fire-area or smoke barrier; not concealed; not open to a Group I or R corridor ; not open to a corridor on nonsprinklered floors; separated from other floor openings by shaft construction	Conditional	Do not open a two-story amenity void into an R corridor	Verified
712.1.13.1	Horizontal fire door	FRR ≥ the assembly penetrated; NFPA 288; labeled	Floor fire doors in rated horizontals	None stated	Conditional	Match hatch ratings to the floor	Verified
712.1.15	Skylights in rated roofs	Unprotected skylights not permitted where the roof must be rated per 705.8.6	Skylights and roof penetrations in a roof required to be rated by 705.8.6	Supporting construction still rated	External verification	Protect or omit skylights on roofs that 705.8.6 requires to be rated	Verified

13. Shaft enclosures and chutes

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
713.4	Shaft fire-resistance rating	Not less than 2 hours where connecting four stories or more ; not less than 1 hour where connecting less than four stories . Story count includes basements, excludes mezzanines . Rating not less than the floor penetrated and need not exceed 2 hours . Nonsymmetrical walls per 703.2.1.1	Shafts protecting floor/roof openings (not interior exit stairs — those are 1023)	Commentary 403.2.1.2 1-hour reduction is not Chapter 7 code and is not adopted	Direct	Specify 2-hour elevator, MEP and refuse shafts for the high-rise; count basements	Verified
713.6	Exterior wall as shaft	Exterior wall that is part of a shaft complies with 705; shaft FRR does not apply to that wall	Exterior wall forming part of a required shaft	Overlay still required for 1021.2 / 1023.7 / 1027.6 (hours not imported)	Conditional	Do not add shaft hours onto an exterior wall that only needs Table 705.5	Verified
713.7	Shaft openings	Protect as fire-barrier openings (716); doors self- or automatic-closing by smoke detection (716.2.6.6)	Openings in a shaft enclosure	Openings other than those necessary for the shaft are prohibited (713.7.1)	Direct	Use smoke-activated shaft doors; do not put borrowed lights into shafts except as required for the shaft purpose	Verified
713.11	Shaft bottom	Enclose at the lowest level with FRR ≥ the lowest floor penetrated and ≥ the shaft; or terminate in a related-use room with FRR and protectives ≥ the shaft; or listed fire dampers at the lowest floor	Shaft that does not extend to the bottom of the building	Exc 1: bottom-only openings plus draftstop or sprinklers. Exc 2: waste/linen → 713.13.4. Exc 3: no combustibles and no interior openings	Conditional	Close elevator/MEP shafts at transfer levels with a 2-hour soffit or a rated equipment room	Verified
713.12	Shaft top	Extend to roof sheathing/deck/slab with Table 601 roof construction; or close below the roof with FRR ≥ topmost floor penetrated and ≥ the shaft; or extend past the roof per 1511	Top of every shaft	713.12.1: no fire/smoke damper required where the shaft goes through the roof into a 1511 rooftop structure with ducts connected directly to HVAC	Direct	Cap shafts at a 2-hour lid or carry them to the roof assembly	Verified
713.13.3	Chute access room	≥ 1-hour fire barriers and/or horizontals; openings ≥ ¾- hour ; doors self- or automatic-closing on smoke; room configured so the room door latches with the chute panel in any position	Waste, recycling or linen chute access	In-unit chute serving and contained in one dwelling unit is exempt (713.13 Exc)	Conditional	Provide a 1-hour chute vestibule; do not open chutes directly to corridors (713.13.1)	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
713.13.4	Chute discharge room	Enclosure FRR not less than the shaft; opening protectives equal to the shaft protection; smoke-activated closing; waste chute shall not terminate in an incinerator room	Chute discharge	Collection rooms without chutes → Table 509.1 (not imported)	Conditional	Match the discharge-room rating to the 2-hour chute shaft	Verified
713.13.6	Chute sprinklers	Approved automatic sprinklers per 903.2.11.2	Waste, recycle and linen chutes	Design density not in this chapter	External verification	Show chute sprinklers on the fire-protection drawings; do not import 903.2.11.2 spacing	Verified
713.14	Elevator hoistways	Construct per 712, 713 and Chapter 30	Elevator, dumbwaiter and other hoistways	Chapter 30 lobby/opening rules not imported	External verification	Keep hoistway construction on 713.4; resolve lobbies in Chapter 30	Verified

14. Penetrations

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
714.4.1.2	Wall through-penetration firestop	ASTM E814 / UL 1479 at not less than 2.49 Pa ; F rating ≥ wall FRR	Through penetrations of fire-resistance-rated walls	T rating not required for walls in this clause. Metallic concrete/masonry path: max 150 mm diameter and 0.1 m² opening with grout full thickness	Direct	Specify F-rated firestop systems matching the wall hour	Verified
714.4.2 Exc 1	Unlisted steel electrical boxes	Each box ≤ 0.01 m² ; aggregate ≤ 0.065 m² in any 10 m² of wall; annular space ≤ 3.2 mm ; opposite-side boxes separated ≥ 600 mm horizontally (or by cavity depth / fireblock / putty pads / listing)	Membrane penetrations of walls/partitions max 2 hours	Listed boxes, putty pads, or Exc 6 for oversized boxes. Do not use commentary 9 m²	Direct	Limit back-to-back boxes; keep the 10 m² aggregate, not the commentary 9 m²	Verified
714.5.1.2	Floor through-penetration firestop	Pressure ≥ 2.49 Pa ; F and T ratings ≥ 1 hour and ≥ floor FRR	Horizontal through penetrations	T not required in a wall cavity; drains in concealed horizontals; metal conduit ≤ 100 mm into metal-enclosed switchgear	Direct	Use F/T firestops through unit and corridor floors	Verified
714.5.1 Exc 1	Multi-floor metallic penetrant	Diameter ≤ 150 mm and aggregate ≤ 92,900 mm² per 10 m² of floor	Steel/ferrous/copper/concrete/masonry through a single rated floor, unlimited floors if both limits met	Concrete-floor grout path similar with ≤ 92,900 mm² per floor	Conditional	Cap unlisted metallic stacks at the aggregate area	Verified
714.5.2 Exc 1–2	Ceiling membrane penetrations	Metallic/concrete/masonry aggregate ≤ 64,500 mm² per 10 m² of ceiling. Steel electrical boxes ≤ 10,300 mm² each, aggregate ≤ 64,500 mm² per 10 m² , annular ≤ 3.2 mm , in max 2-hour horizontals	Membrane penetrations of rated floor/roof-ceiling assemblies tested without penetrations	Listed boxes and firestop exceptions	Direct	Coordinate downlights and boxes with the 64,500 mm² / 10 m² ceiling cap	Verified
714.5.4	Smoke-barrier penetration leakage	UL 1479 air leakage at 75 Pa : ≤ 1.5 m³/min per m² of opening, or cumulative ≤ 25 L/s per 9 m² of wall or floor	Penetrations in smoke barriers	Ambient and elevated tests	Conditional	Require L-rated firestops on AOR/lobby smoke barriers	Verified
714.6.1–714.6.2	Nonrated floor penetrations	Noncombustible penetrants connecting max 5 stories with noncombustible or classified fill; any penetrants connecting max 2 stories with approved annular fill	Non-fire-resistance-rated floors	None stated	Conditional	Shaft nonrated transfer floors still need annular fill or a 713 enclosure	Verified

15. Joints and voids

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
715.3	Fire-resistant joint system	Duration ≥ the assembly FRR (ASTM E1966 / UL 2079 as applicable)	Joints in or between rated walls, floors and roofs	Ten listed location exceptions; control joints ≤ 16 mm tested as part of E119/UL 263	Direct	Specify listed joint systems at floor/wall and wall/wall intersections	Verified
715.3.1 Exc	Exterior joint fire exposure	Where FSD > 3000 mm , test from the interior only	Nonsymmetrical wall joints	Else both faces; shortest duration	Conditional	Two-sided joint listings on close lot-line walls	Verified
715.4	Curtain-wall perimeter containment	ASTM E2307 F rating ≥ floor FRR	Void between a curtain wall and a rated floor	Vision glass extending to the finished floor: approved material, 2.5 Pa , time ≥ floor FRR	Direct	Provide an E2307 safing line at every curtain-wall floor edge	Verified
715.8	Smoke-barrier joint leakage	≤ 7.75 L/s per m at 75 Pa , ambient and elevated (UL 2079)	Joints in smoke barriers and curtain-wall/horizontal smoke-barrier voids	None stated	Conditional	Specify L-rated joints on smoke-barrier lines	Verified

16. Opening protectives

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
716.2.2.1	Corridor / smoke-barrier fire door	Minimum 20 minutes ; NFPA 252 / UL 10C without hose stream	Table 716.1(2) 20-minute doors in corridor or smoke-barrier walls	Viewport ≤ 25 mm diameter with ≥ 6 mm glass disc in a holder that will not melt at 927°C	Direct	Specify 20-minute smoke-and-draft corridor doors; hose-stream not required on this path	Verified
716.2.2.1.1	Smoke and draft control	Leakage ≤ 15 L/s per m² at 25 Pa (UL 1784); louvers prohibited	Corridor and smoke-barrier fire doors	Terminated stops prohibited where cited	Direct	Put UL 1784 / “S” label on corridor unit-entry and lobby doors	Verified
716.2.2.2	Other 0.5-hour partition doors	Minimum 20 minutes with hose stream (NFPA 252 / UL 10B / UL 10C)	Fire partitions 0.5 hour other than corridors	Type I/II unit walls are 1 hour, so this row is the ½-hour exception path	Conditional	If ½-hour unit walls are used (Type IIB/IIIB/VB + 13), test those doors with hose stream	Verified
716.2.2.3	Exit-door temperature rise	Maximum transmitted temperature rise not more than 250°C above ambient at 30 minutes	Fire doors in interior exit stairways, ramps and exit passageways	Not required where sprinklers throughout are 903.3.1.1 or 903.3.1.2	Conditional	Use 250°C from the code text (not commentary 232°C) unless the sprinkler exception is locked; Table 716.1(1) “T” mark is 232°C / 30 min	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
716.2.2.3.1	Exit-door vision panel	Fire-protection-rated glazing > 0.065 m² not permitted ; larger panels only if fire-resistance-rated and meeting 716.2.2.3 temperature rise	Glazing in those exit doors	None stated	Direct	Keep stair-door vision panels ≤ 0.065 m² unless W-rated glazing is listed for the door	Verified
Table 716.1(2)	Fire-wall / >1 h fire-barrier doors	Wall 4 h → door 3 h , sidelights NP, W-240 ; wall 3 h → door 3 h (note d: two 1½ h doors ≡ one 3 h), W-180 ; wall 2 h → door 1½ h , vision 64,500 mm² (≤ D-H-90 , > D-H-W-90), sidelights NP, W-120 ; wall 1½ h → door 1½ h , same vision split, W-90	Fire walls and fire barriers with required FRR greater than 1 hour	Remainder of the table (shafts, 1 h barriers, partitions, corridors) is flattened — do not reconstruct from memory	Conditional	Use this block for 2-hour shafts/fire areas; hold all other Table 716.1(2) rows as Verify source	Verified
Table 716.1(2) cont.	Exterior-wall and smoke-barrier doors	Exterior: wall 3 h → door 1½ h , vision 64,500 mm² D-H-90 , SL NP, W-180 ; wall 2 h → door 1½ h , max size tested, D-H-90 or D-H-W-90 ; wall 1 h → door ¾ h, D-H-45 . Smoke barrier: wall 1 h → door ½ h , max size tested, D-20	Exterior walls and smoke barriers	Note h: fire-protection-rated glazing not permitted in SBC 801 §1207 ESS barriers	Conditional	Match exterior opening-protective hours to Table 705.5 wall hours	Verified
Table 716.1(2) remainder	Shaft, 1 h barrier, partition, corridor door ratings	Flattened repeat of 21164,500 mm² D-H-90 after the double-fire-wall 321½ token	Other assembly types in Table 716.1(2)	Do not fill from IBC memory	External verification	Verify the published table for 1-hour shaft, corridor and unit-entry door hours before issuing the door schedule	Verify source
Table 716.1(3)	Fire-window ratings	Fire walls: NP. Fire barriers 1 h : NP (general). Atrium / incidental / mixed (707.3.6 / 707.3.7 / 707.3.9): wall 1 h → window ¾ h, OH-45 or W-60 . Fire partitions 0.5 h : ½ h, OH-20 or W-30 . Smoke barriers 1 h : ¾ h, OH-45 or W-60 . Exterior 1 h : ¾ h ; exterior 0.5 h : ½ h . Party walls: NP	Fire-protection-rated windows	Fire-barrier > ch1 token and exterior > 1 → 1/2 window hour are OCR holds	Conditional	Prefer W-rated glazing in 2-hour shafts; use OH-45 only where the 1-hour mixed/incidental/atrium row applies	Verified
Table 716.1(3) OCR	Fire-barrier >1 h and exterior >1 h windows	Wall token > ch1 : NP. Exterior > 1 window 1/2 (1½ vs ½ unresolved)	Those table rows	Do not adopt ½ hour from the flattened exterior cell	External verification	Verify the published Table 716.1(3) before specifying windows in >1-hour exterior walls	Verify source
716.2.6.6	Smoke-activated closing delay	Closing shall begin not more than 10 s after smoke-detector actuation	Hold-open automatic-closing doors at listed locations	None stated	Conditional	Set hold-open release delay ≤ 10 s	Verified
716.3.2.1.2	Interior fire-window area	Fire-protection-rated window assemblies ≤ 25% of the common wall with any room	Interior fire-protection-rated windows	Fire barriers also limited by 707.6 length	Conditional	Cap borrowed-light fire windows at 25% of the room's common wall	Verified
716.3.4	Unrated exterior protected openings	≥ ¾-hour fire-protection rating	Non-FRR exterior walls that still require protected openings under 705.3 / 705.8 / 705.8.5 / 705.8.6	0.5-hour partitions may use 0.33-hour fire-protection glazing	Conditional	Where Table 705.8 requires protected openings in a 0-hour wall, specify ¾-hour protectives	Verified

17. Ducts and air-transfer openings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Table 717.3.2.1	Fire-damper rating	Assemblies < 3 hours : damper ≥ 1.5 hours . Assemblies 3 hours or greater : damper ≥ 3 hours	Fire dampers (and combination dampers per 717.3.2.3)	None stated	Direct	Use 1.5-hour dampers in 1- and 2-hour walls; 3-hour dampers only in 3-hour fire walls	Verified
717.3.2.2	Smoke-damper class	Class I or II ; elevated-temperature rating ≥ 121°C	Smoke dampers	None stated	Direct	Specify Class I or II smoke dampers at ≥ 121°C	Verified
717.3.3.1	Fire-damper actuation	Operating temperature ≈ 10°C above normal duct temperature, not less than 71°C ; not more than 177°C if in a Section 909 smoke-control system	Primary heat-responsive device	None stated	Direct	Select fusible links at ≥ 71°C	Verified
717.3.3.2	Smoke-detector distance	In-duct detector or sampling tubes within 1500 mm of the damper, no inlets/outlets between; transfer-opening spot detector within 1500 mm horizontally	Smoke-damper actuation	Corridor detection or area detection alternatives 4–5	Direct	Place duct detectors within 1500 mm of each smoke damper	Verified
717.4.1–717.4.2	Damper access	Access door not less than 300 mm square, or a removable duct section, where fusible links or internal operators exist; permanent label letters not less than 12.5 mm ("FIRE/SMOKE DAMPER", "SMOKE DAMPER" or "FIRE DAMPER")	Fire and smoke dampers	Remote inspection per NFPA 80/105 where access is restricted	Direct	Show 300 mm access doors and labels on MEP drawings	Verified
717.5.2	Fire-barrier duct dampers	Listed fire dampers	Ducts and air-transfer openings in fire barriers	Do not penetrate exit enclosures except as 1023.5 / 1024.6 permit. Exc 3: omit FD where wall ≤ 1 h , steel ≥ 0.47 mm , 13 or 13R, not Group H, fully ducted HVAC	Direct	Provide fire dampers at 2-hour occupancy/fire-area barriers unless a numbered exception applies	Verified
717.5.3	Shaft fire and smoke dampers	Listed fire and smoke dampers at permitted shaft penetrations	Shaft ducts and air-transfer openings	Exc 1.1: omit FD for steel subducts ≥ 0.47 mm extending ≥ 550 mm with continuous 909.11 exhaust. Exc 2 (Group B and R, 903.3.1.1 throughout): omit smoke dampers on kitchen/dryer/bath/toilet exhaust with the same subduct/ 550 mm /continuous fan. Exc 1.4 / 3: garage shafts separated ≥ 2 h . Exc 5: FD prohibited on kitchen/dryer exhaust per SBC 501	Direct	Combine Exc 1.1 and Exc 2 to omit both dampers on R toilet/kitchen exhaust only if NFPA 13 (not 13R) is locked	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
717.5.4	Fire-partition duct dampers	Listed fire dampers	Fire-partition penetrations, other than Group H	Exc 4: omit FD for ducted HVAC, wall ≤ 1 h , steel ≥ 0.47 mm , 13 or 13R. Corridor dampers per 717.5.4.1	Direct	Provide fire or corridor dampers at unit/corridor walls unless the ducted-HVAC exception is documented	Verified
717.5.5	Smoke-barrier duct dampers	Listed smoke damper in each duct or air-transfer opening	Smoke barriers	Steel ducts opening to one smoke compartment only; I-2 Condition 2 QR-sprinkler exception	Conditional	Smoke-damp AOR and elevator-lobby smoke-barrier ducts	Verified
717.6.1 Exc	Three-floor residential duct	Max 3 floors ; steel ≥ 0.47 mm in a wall cavity; contained in one dwelling/sleeping unit; diameter ≤ 100 mm ; aggregate ≤ 0.065 m² per 9 m² of floor; 2.5 Pa annular; ceiling-radiation dampers at rated-ceiling grilles	Dwelling-unit ducts without a fire damper at each floor	This 9 m² token is CODE (unlike 714.4.2 10 m²)	Conditional	Use only for in-unit stacks of three floors or fewer	Verified

18. Concealed spaces, plaster and insulation

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
718.2.1–718.2.2	Combustible-wall fireblocking	Materials include 50 mm nominal lumber, two 25 mm , 18 mm WSP backed 18 mm , 20 mm particleboard backed 20 mm , 12.5 mm gypsum, 6 mm cement millboard, or mineral wool; horizontal intervals not more than 3000 mm ; unfaced fiberglass fill ≥ 400 mm high	Combustible concealed stud, furred or staggered walls	Type I/II combustible concealed uses limited by 718.5	Conditional	Fireblock combustible podium or Type III–V walls at 3 m horizontally and at every floor	Verified
718.2.6	Combustible exterior-covering fireblock	Intervals not more than 6000 mm either way; concealed area not more than 9 m² ; noncontinuous closed ends ≥ 100 mm apart	Combustible exterior coverings/frames (Section 1405)	Metal covering skip: aluminium ≥ 0.5 mm or steel ≥ 0.4 mm on noncombustible framing, or NFPA 285 covering	Conditional	Fireblock rainscreen cavities at 6 m / 9 m² unless a metal/NFPA 285 exception applies	Verified
718.3	Non-R floor draftstopping	Subdivide combustible floor/ceiling areas not more than 100 m²	Occupancies other than Group R	Group R follows 708.4.2 . Omit if 903.3.1.1 throughout	Not typical	R floors use 708.4.2, not this 100 m² cap	Verified
718.4	Non-R attic draftstopping	Subdivide combustible attic/roof concealed areas not more than 280 m²	Occupancies other than Group R	Group R follows 708.4.2 . Omit if 903.3.1.1. Ventilate 1202.2.1	Not typical	R attics use 708.4.2; the high-rise ≤ 18 m / 4-storey reduction does not apply	Verified
718.3.1	Draftstop materials	≥ 12.5 mm gypsum, ≥ 10 mm wood structural panel, ≥ 10 mm particleboard, ≥ 25 mm nominal lumber, or listed mineral/glass fibre	Floor draftstops	Attic materials 718.4.1	Conditional	Specify gypsum or wood-panel draftstops where 708.4.2 requires them	Verified
719.2–719.5	Plaster fire-resistance equivalents	12.5 mm unsanded gypsum ≡ 20 mm 1:3 gypsum-sand ≡ 25 mm portland-cement sand. Plaster > 25 mm needs extra lath ≥ 20 mm from the outer surface. 12.5 mm plaster may replace 12.5 mm concrete cover if ≥ 10 mm concrete remains on floors and ≥ 25 mm on columns	Plaster used for fire resistance	None stated	Conditional	Do not substitute finish plaster for tested SFRM on steel	Verified
720.2–720.3	Insulation flame spread	Concealed or exposed: flame-spread index ≤ 25 and smoke-developed index ≤ 450 (ASTM E84 or UL 723)	Thermal- and sound-insulating materials	Cellulosic loose-fill: no FS limit if SD ≤ 450 by CAN/ULC S102.2 (720.6). Foam → Chapter 26. Plenums → SBC 501	Direct	Specify FS 25 / SD 450 insulation in walls, floors and attics	Verified
720.3.1	Exposed attic-floor insulation	Critical radiant flux not less than 1200 W/m² (ASTM E970)	Exposed insulation on attic floors	None stated	Conditional	If attic-floor insulation is left exposed, require the E970 flux	Verified
720.7	Pipe insulation	Flame-spread ≤ 25 , smoke-developed ≤ 450	Pipe and tubing insulation and covering	Plenums comply with SBC 501 (values not imported)	Direct	Keep pipe lagging on the FS 25 / SD 450 path except in plenums	Verified

19. Prescriptive and calculated fire resistance

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
721.1	Prescriptive assemblies	Materials in Tables 721.1(1), 721.1(2) and 721.1(3) are assumed to have the tabulated FRR; added heat-dissipation-changing materials need data that the rating is not reduced	Alternative to a tested listing	Tables are OCR-destroyed — no cell adopted	External verification	Prefer UL/GA listings. Do not reconstruct brick, gypsum or deck thicknesses from memory	Verify source
721.1.3	Cast-in-place column ties	Wire not less than 4.5 mm diameter, spiral pitch not more than 200 mm , or equivalent	Cast-in-place concrete protection of steel columns	None stated	Conditional	If using 721.1(1) column encasement, detail the spiral ties	Verified
721.1.5	Variable tendon cover	Average cover not less than Table 721.1(1); no tendon less than one-half of the table; absolute minimum 20 mm slabs / 25 mm beams; reduced-cover tendons contribute ≤ 50% of required (M_u) if member < 0.23 m² , ≤ 65% if larger	Prestressed members with unequal cover	Table 721.1(1) cover values themselves are Verify source	Conditional	Verify published Table 721.1(1) before using calculated tendon cover	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
722.2–722.6	Calculated concrete, masonry, steel, wood	Code-text limits that are readable include foam insulation < 25 mm disregarded / ≥ 25 mm (R_n = 5) minutes; exterior one-sided fire FSD > 1500 mm (concrete/masonry) or > 3000 mm (wood); column (f'_c) > 83 MPa min dimension 600 mm ; R/C column cover 25 mm × hours cap 50 mm ; substitute-beam SFRM thickness ≥ 10 mm ; calculated wood assemblies not more than 1 hour	Calculated fire resistance in lieu of a test	Equations 7-3/7-5, 7-12, 7-14 (Ro), 7-16 and 7-17 (h1) are OCR-unresolved. All 722.* tables are Verify source	External verification	Do not run 722 calculations from this extract; use a published ACI 216.1 / AISI / AWC reference after source check	Verified
722.7.2.1	Interior mass-timber gypsum fastening	Type S screws penetrate ≥ 25 mm into mass timber; base and subsequent layers 300 mm o.c. both ways; subsequent layer offset 100 mm ; panel edges offset 450 mm ; edge screws 25 mm min / 50 mm max ; 32 mm corner bead at 602.4.2.2 junctions	Type IV-A/B/C interior noncombustible protection	Third layer may use 25 mm No. 6 Type S to AISI S220 furring. Table 722.7.1(1)/(2) contribution minutes concatenated (1402803, 12.5 mm / 16 mm) — not adopted	Conditional	If mass timber is specified, use these fastening rules and verify published protection minutes	Verified
722.7.2.2	Exterior mass-timber gypsum fastening	Field 300 mm o.c. each way; joints/ends 150 mm o.c.; edge fasteners 25–50 mm ; nails min 12 gage, 11 mm head, penetrate ≥ 25 mm ; screws ASTM C1002 penetrate ≥ 25 mm	Exterior side of mass timber	None stated	Conditional	Detail exterior gypsum on mass timber to these spacings	Verified

20. Project-use controls

1. Use **Verified** rows for coordination after the row trigger and branch are confirmed.
2. Treat every **Verify source** row (Figure 705.7, flattened Table 716.1(2) remainder, Table 716.1(3) OCR tokens, and all Tables 721.1 / 722.*) as a design hold point; no affected value is to be placed on issued-for-approval drawings without a published-source check.
3. Do not import Table 601, 508.4, 509.1, 1020.2, 403.2.1.2, 403.2.3, Chapter 9 densities, Chapter 10 corridor hours, Chapter 14 cladding limits, NFPA 82, or SBC 501 plenum values from memory, CS.md, or commentary.
4. Do not treat NFPA 13R as the Table 705.8 UP, S branch or as 708.3 Exc 2 / 711.2.4.3 Exc / 717.5.3 Exc 2.
5. Do not apply Group R-3 / U opening, parapet or projection exceptions, I-2/I-3 smoke-compartment walls, mall tenant partitions, or the R-2 **≤ 4** storey / **≤ 18 m** attic-draftstop reduction to this high-rise.
6. Keep CODE **250°C** (716.2.2.3) distinct from Table 716.1(1) **232°C** “T” marking and from commentary 232°C. Keep 714.4.2 aggregate **10 m²**, not commentary **9 m²**.
7. Record construction type, FSD by elevation, sprinkler standard, fire-wall vs fire-area strategy, and shaft storey counts in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped fire-protection compliance.

21. Coverage summary

Internal inventory of the attached Chapter 7 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 2863
- **Verified:** 517
- **Verify source:** 2346

Counts by top-level section

Top-level section	Records
701	0
702	0
703	7
704	19
705	102
706	54
707	28
708	17
709	4
710	5
711	12
712	14
713	25
714	31
715	6
716	48
717	30
718	18

Top-level section	Records
719	5
720	6
721	1288
722	1144

Section 721 = 8 readable code-text records + ~1280 Verify-source cells in Tables 721.1(1)–(3). Section 722 = 88 readable code-text records + 6 OCR-unresolved equations + ~1050 Verify-source cells in Tables 722.*. Table-cell counts for 721/722 are estimates of independently checkable cells that cannot be adopted.

Appended-table coverage

Appended table	Records	Verify source records
Table 705.2	4	0
Table 705.5 + numeric notes	24	0
Table 705.8 + numeric notes	29	0
Table 706.4 + note a	5	0
Table 707.3.10	4	0
Table 716.1(1)	1	0
Table 716.1(2)	14	2
Table 716.1(3)	10	2
Table 717.3.2.1	2	0
Figure 705.7 (F_{eo})	1	1
Tables 721.1(1)–(3)	1280	1280
Tables 722.2 through 722.7 (printed titles include 22.5.1(6), 72.5.1(10), 72.6.2(*), 72.7.1(*))	1050	1050

No Chapter 7 *_CS.md was found. Commentary figures and commentary calculations (ASTM E136 furnace temperatures, 9 m² outlet-box commentary, 232°C commentary vs 250°C code, 403.2.1.2 128 m shaft reduction) were not inventoried as requirements.

22. Unresolved-source register

Hold points for the **2346** Verify-source inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
Figure 705.7 (F_{eo})	Normative figure; no numeric lookup table in the extract	Do not compute equivalent protected area (Eq. 7-1) until the published figure is read
Table 716.1(2) remainder	After the unique 4/3/2/1½-hour fire-wall/fire-barrier block, cells repeat 21164,500 mm² D-H-90 ; double-wall row OCR 321%	Verify published shaft, 1-hour barrier, partition and corridor door hours before issuing the door schedule
Table 716.1(3) fire-barrier >ch1	Wall-rating token unreadable	Treat general fire-barrier windows as NP until the published >1-hour row is confirmed
Table 716.1(3) exterior >1 → 1/2	Likely 1½-hour window OCR as ½	Do not specify ½-hour fire windows in >1-hour exterior walls from this extract
Tables 721.1(1), 721.1(2), 721.1(3)	Concatenated HTML; titles OCR as 7.1.12(a), 7.12(2), 7.1.13, 21.13 ; thickness cells unreadable (~1280 records)	Do not reconstruct prescriptive assembly thicknesses. Use tested listings
Tables 722.2.1.1 through 722.7.1(2)	Flattened W/D, cover, equivalent-thickness and mass-timber-minute grids; titles OCR as 22.5.1(6), 72.5.1(10), 72.6.2, 72.7.1 (~1050 records)	Do not adopt concatenated 1402803 or 12.5/16 mm Type X minutes. Verify published 722 tables if calculated FRR is used
Equations 7-3/7-5, 7-12, 7-14 (R_o), 7-16, 7-17	Formula parentheses and (h_1) multiplier destroyed in OCR (6 records)	Do not run those calculations from this extract